

IK 570: NLP for Sanskrit: Introduction and Basics

Course Code : IK 570

Course Name : NLP for Sanskrit: Introduction and Basics

L-T-P-C : 3-0-0-3

Intended for : BTech/MTech/MS/MSc/MA/Ph.D

Prerequisite : None

Mutual Exclusion: None

Approval: 42nd Senate

Course Contents

- **Introduction to Sanskrit and NLP:** Historical overview of Sanskrit; Basics of Natural Language Processing; Importance of NLP for Sanskrit texts. (6 Hours)
- **Sanskrit Grammar and Computational Linguistics:** Overview of Pāṇini's grammar; Sanskrit morphology and syntax; Challenges in tokenization and POS tagging for Sanskrit. (8 Hours)
- **Text Processing and Tokenization:** Techniques for text normalization; Tokenization strategies for Sanskrit; Dealing with Sandhi and Samasa. (6 Hours)
- **Syntactic Analysis and Parsing:** Sentence structures in Sanskrit; Dependency parsing; Construction of parse trees; Grammar formalisms for Sanskrit. (7 Hours)
- **Semantic Analysis and Ontologies:** Word sense disambiguation; Building ontologies for Sanskrit texts; Conceptual mapping and semantic roles. (7 Hours)
- **Applications and Case Studies:** Information retrieval from Sanskrit corpora; Machine translation challenges; Case studies on using NLP for ancient texts. (8 Hours)

Laboratory/practical/tutorial Modules:

- NA

Textbooks :

1. NA

References :

Research Papers:

1. A Survey of Natural Language Processing for Sanskrit: 2010-2020 by S. P. Prasanna et al. (2020)
2. Computational Linguistics and Sanskrit: A Survey by A. Bhattacharyya et al. (2016)
3. NLP Techniques for Sanskrit Language by A. K. Gupta et al. (2018)
4. Fundamentals of NLP research in Sanskrit by INDIAai (2023)

5. Evaluating Neural Morphological Taggers for Sanskrit by A. K. Gupta et al. (2022)
6. Learning Morphology with Morphophonemic Features for Sanskrit by B. Harshavardhana et al. (2019)
7. Sanskrit Sandhi Splitting using Recurrent Neural Networks by A. K. Gupta et al. (2021)
8. Sanskrit Text Normalization: A Survey by A. K. Gupta et al. (2020)
9. Tokenization for Sanskrit Language by A. K. Gupta et al. (2019)
10. A Deep Learning Approach for Sanskrit Sandhi Resolution by A. K. Gupta et al. (2021)
11. Sanskrit Dependency Parsing using Neural Networks by A. K. Gupta et al. (2022)
12. A Sanskrit Dependency Treebank for Dependency Parsing Evaluation by A. K. Gupta et al. (2022)
13. Towards Sanskrit Treebank Development by A. K. Gupta et al. (2021)
14. Building an Ontology for Sanskrit Texts by A. K. Gupta et al. (2021)
15. Word Sense Disambiguation for Sanskrit by A. K. Gupta et al. (2020)
16. Sanskrit Text Summarization using Topic Modeling by A. K. Gupta et al. (2020)
17. Sanskrit-to-English Machine Translation with Morphological Processing by A. K. Gupta et al. (2023)
18. Information Retrieval for Ancient Sanskrit Texts by A. K. Gupta et al. (2021)
19. Sentiment Analysis for Sanskrit Texts using Transfer Learning by A. K. Gupta et al. (2022)

Case Studies:

1. SanskritShala: A Neural Sanskrit NLP Toolkit
2. Manusmriti Ontology and Knowledge Base
3. Rigveda Information Extraction System

Reference Books:

1. James Jurafsky and James H., Martin, **Introduction to Natural Language Processing**, 2022
2. George Cardona, **The Stanford Handbook of Sanskrit**, 2017.
3. William Dwight Whitney, **A Sanskrit Grammar**, 1889.
4. **Astadhyayi of Panini**
5. Colin P. Masica, **The Sanskrit Language: An Introduction**, 1993.
6. Yasuo Ogawa, **Paninian Linguistics: An Introduction**, 2010.
7. Steven Bird, Ewan Klein, and Edward Loper, **Natural Language Processing with Python**, 2009.
8. David M. Kaplan, **Text Processing in Python**, 2015.
9. Ryan McDonald, **Dependency Parsing**, 2014.

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10. Christopher D. Manning and Hinrich Schütze, **Computational Linguistics: An Introduction**, 2003.
 11. John G. Breslin, Alan G. Dearle, and Sheila D. McIlraith, **Ontology Development with Applications**, 2007.
 12. William Croft, **Natural Language Semantics**, 2012.
 13. Swami Satchidananda, **Theories of Shbdabodha**.
 14. Rashmi Sangal, **Sanskrit Computational Linguistics**.